Quran-Tajzee: Balanced Quran Divisions that End Where a Thought Ends

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***Abstract*—The Madina Mushaf divides the Quran into 30 juz, 60 hizb and 240 quarters (rub^{c}), taken from al-Ṣafāqusī's *Ghayth al-Naf^{c}*. His stated rule is that each quarter ends at a complete stop, at a position that is agreed or well known; the quarters are not computed. We measure the transmitted divisions on the Tanzil text with two length models, Hafs recitation in harakat with madd lengths and DigitalKhatt print width, which correlate at $r = 0.990$ per aya. The juz lie within 2.4% of equal-letter positions, but the quarters lie 10× further from equal positions than aya ends force and vary 2.2× in length; and 90 of 239 boundaries fall inside a subject phrase. Keeping the traditional juz, we re-cut hizb and quarters by an exact dynamic program restricted to the starts of 1,172 reviewed subject phrases of 1–10 ayat. The worst hizb drops from ±13.9% to ±3.6% of half a juz (CV 6.9% → 1.4%), and no new boundary falls inside a phrase. For the quarters, a text-checked review of 137 boundaries accepts 60 phrase splits at complete stops and narrows the spread: CV 11.7% → 6.8%, longest/shortest 2.19× → 1.57×; what remains uneven lies inside long narratives, where no closer stop exists.**

***Index Terms*—Quran, text segmentation, partitioning, dynamic programming, typesetting, recitation, tajweed**

I. Introduction

Readers, teachers and printers organise the Quran by its divisions: a juz a day completes it in a month, a hizb or a quarter is a daily portion or a unit of revision, and printed Mushafs mark every quarter. Those divisions are of scholarly judgement (*ijtihādī*), not revelation, which is why mushafs and regions differ on them [3]. The Madina Mushaf takes its 30 juz, 60 hizb and their quarters from *Ghayth al-Naf^{c}* by al-Ṣafāqusī (d. 1118 AH) [1], [2].

Two properties are wanted of a division: units of similar length, and units that are complete in meaning. The juz satisfy the first; the quarters satisfy neither fully (Section II). This paper (i) measures the traditional divisions against their own stated method, (ii) defines a length model for recitation and print, (iii) solves the hizb and quarters inside the traditional juz exactly, with every boundary at the start of a subject phrase, and (iv) shows how targeted phrase splits, each checked against the text, reach the balance that aya granularity allows. Data, code and every number in this paper are in the open repository [15].

II. The traditional method and its outcome

In the terminology section of his introduction al-Ṣafāqusī states his method [1]: he does not compute boundaries but follows only the agreed or well-known ones, "ولا أمشي إلا على المتفق عليه أو المشهور"; he treats each quarter as a unit, "وأذكر حكم كل ربع بانفراده"; and he closes each on a proper stop, naming its last word, the kind of stop it allows and whether it is an aya end. At 2:25 he writes "خالدون تام في أعلى درجاته وفاصلة ومنتهى الربع بإجماع": a complete stop of the highest degree, an aya end, the end of the quarter by consensus. Where traditions differ he lists the positions of the Maghāriba and the Mashāriqa and states a preference; the two cases we checked both match the Madina Mushaf (45:12, 82:1).

Drawn at stops, not by count. The early divisions were made by counting letters [4], [5]. Counting the 325,386 written letters of the text (basmala excluded), the traditional juz start on average 2.4% of a juz from equal-letter positions. Inside each juz, however, the quarters start 15% of a quarter from equal positions, while the nearest aya end is only 1.5% away: 10× further than aya granularity forces. Quarter lengths vary 2.2× in letters. The quarters were placed at the ends of thoughts, as the method says, not by count.

But not always at the ends of thoughts. We segmented the text into subject phrases (Section III). Against that reading, 90 of the 239 traditional boundaries, 79 of the 210 hizb and quarter boundaries inside the juz, fall inside a phrase, at a median 50% of the way through it. Some open mid-dialogue or mid-story: juz 16 at "قال ألم أقل لك" (18:75), juz 27 at "قال فما خطبكم أيها المرسلون" (51:31), a hizb at "فنبذناه بالعراء" (37:145). The quarters thus pay the price of uneven length without fully keeping the rule they were drawn by. This is the problem we address.

III. Data and length model

Text. Tanzil Uthmani 1.1, 6,236 ayat in the Kufan count [10].

Subject phrases. The text was divided into runs of 1–10 consecutive ayat on one subject, drafted by reading the text and reviewed by hand: 1,094 phrases as a reading of the text, 1,172 after the review round of Section IV. A phrase start is our proxy for a complete stop: the next aya does not continue the previous subject.

Recitation size. Per aya, the phonemes of the Hafs recitation from *tajweed-embeddings* [12]: a madd letter counts its length in harakat (2, 4 or 6, with *tawassuṭ* for the optional lengths), a silent letter 0, every other letter 1. Each aya is recited continuously and stopped at its end. A surah opening adds the basmala (19 harakat), except At-Tawba. The whole text is 398,670 harakat.

Print size. Per aya, the width of the aya set as a single line in the DigitalKhatt Madina font [11], shaped with HarfBuzz by Quran-Nazm [13], with its aya marker and one space.

The two sizes correlate at $r = 0.990$ per aya and $r = 0.994$ per phrase. We mix them 50/50 as shares of their totals, $c_a = \frac{1}{2} r_a/R + \frac{1}{2} p_a/P$.

IV. Method

Keep the juz. An exact solver over the phrases (penalising mid-surah starts) can make the 30 juz more even, but only from ±6.2% to ±5.0%. The traditional juz are kept; the imbalance is in the finer divisions.

Nested exact cuts. Each juz is cut into two hizb, then each hizb into four quarters, the way the divisions nest. A container G of ayat cut into k parts has target $T = |G|/k$ and the admissible cuts $\mathcal{S}$ are phrase starts that follow an agreed aya end (below), outside ayat 2–7 of a surah (juz 28 exempt). For the hizb, lexicographically we

$$\min_{\text{cuts}\subset\mathcal{S}} \max_q \frac{|S_q - T|}{T}, \quad \text{then} \quad \min \sum_q (S_q - T)^2, \quad (1)$$

on the 50/50 mix S. The quarters weigh three terms equally: recitation R, print P (each against its own target) and the pause before each cut j,

$$\min_{\text{cuts}\subset\mathcal{S}} \tfrac{1}{3} \sum_q \Big(\frac{R_q - T_R}{T_R}\Big)^2 + \tfrac{1}{3} \sum_q \Big(\frac{P_q - T_P}{T_P}\Big)^2 + \tfrac{1}{3} \sum_j (v\,\pi_j)^2, \quad (2)$$

where π_j are pause points of the aya end before the cut, from the classical grades (below): 1 for a surah end, 2 complete (تام), 3 sufficient (كاف), 5 weaker (حسن, جائز, or not graded), 9 not a stop (ليس بوقف). One point is worth $v = 5\%$ of a quarter, so a sufficient stop must buy more than 5% of balance to be preferred to a complete one. All 1,172 phrase starts carry their points (`data/rub/phrase-pauses.tsv`). Both objectives are solved by the dynamic program $F_k(i) = \min_{j<i} [F_{k-1}(j) \oplus g(j,i)]$ over prefix sums, with $\oplus = \max$ or $+$ [14]. Contiguous partitioning has optimal substructure, so the solution is the global optimum; tests check it against exhaustive search. A *penalized* variant also admits any aya end at a cost m^2 per mid-phrase start; we use it only to *propose* phrase splits.

Implementation. The solver is plain Python 3 (3.10 or later) using only the standard library (`csv`, `json`, `bisect`, `dataclasses`); it uses no numerical, LP or MIP package, since the problem needs none. A hizb container is solved in two passes of the same DP: the first finds the optimal worst deviation M^*; the second minimises the sum of squares over cuts whose every part stays within M^* (pruned by that bound, tolerance 10^{-12}), which gives the lexicographic optimum of (1). A quarter container needs one pass, since every term of (2) is a sum over parts and cuts. A pass costs $O(k\,n^2)$ for n admissible starts in a container of k parts; the nested solution of all 30 juz takes about a second on one CPU core. Unit tests compare the DP with exhaustive search (`itertools.combinations`) on small inputs. The two length scores are computed once by external tools (tajweed-embeddings and Quran-Nazm with DigitalKhatt [11]–[13]) and stored per aya, so the solver reads only tables. Code and data are open [15].

Review process. Balance is limited by how fine the phrases are, so phrases are refined where the divisions need it, one container at a time. (1) For every aya end in the container, the balance is computed if the next unit started there. (2) The candidates are the aya ends closer to balance than the current phrase-start solution; all lie inside a phrase. (3) Each candidate is read in context and accepted only at a complete stop: the aya before it closes its sentence and subject, and the aya after it does not lean on it. A reply (قال), a pointer back (أولئك, الذين), a ثم/ف continuation, or a description of what precedes it disqualifies a candidate. (4) Among accepted candidates the most even wins; if none qualifies, the current start is kept, however uneven. (5) An accepted candidate splits its phrase, with a topic for each part, so the rule "every boundary opens a phrase" is never relaxed. Every candidate, its balance, the decision and the reason are recorded in a table that a human reviewer approves or overturns; the phrase list is rebuilt from the reading and the accepted rows.

Accepted (juz 10). The second hizb started at 9:29, −3.7% off. Candidate 9:30, −1.5%, lies inside the phrase 9:29–35. The aya before it ends "وهم صاغرون", a complete stop closing the command to fight; 9:30 "وقالت اليهود عزير ابن الله" opens a new subject, the claims of the People of the Book. The phrase is split into 9:29 and 9:30–35.

Rejected (juz 6). The second hizb starts at 5:27, +3.6%. Candidate 5:24, +0.2%, is almost exactly at the middle, but "قالوا يا موسى إنا لن ندخلها" is the people's reply inside the dialogue of 5:20–26; every closer point is a turn of that dialogue. The start is kept at 5:27, the story of the two sons of Ādam, at the cost of 3.4 points of balance.

V. Results

Hizb. One review round checked all 30 second-hizb starts: 10 were already at the best point, 7 phrases were split at complete stops (e.g. 9:30 "وقالت اليهود", 10:28 "ويوم نحشرهم"), and 13 were kept because every closer point lies inside a story or dialogue (e.g. juz 6, +3.6%, inside

Table I: Traditional and new divisions (50/50 size mix). Hizb against $\frac{1}{2}$ of their juz, quarters against $\frac{1}{4}$ of their hizb; ratios over the whole Quran.

	Traditional	Quran-Tajzee
Worst hizb	±13.9%	±3.6%
Hizb CV	6.9%	1.4%
Longest/shortest hizb	1.36×	1.17×
Worst quarter	±38.3%	±20.8%
Quarter CV	11.7%	6.8%
Longest/shortest quarter	2.19×	1.57×
Starts inside a phrase (of 210)	79	0

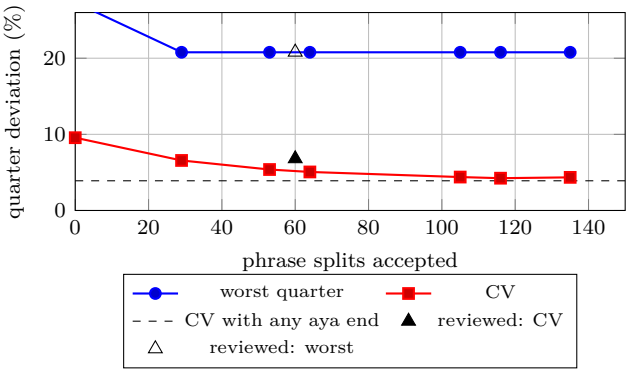


Figure 1: Quarter balance against the number of phrase splits accepted, if every split the solver proposes were accepted (lines), and after the text-checked review (triangles).

the dialogue of Mūsā and his people). The 11 phrases cut by traditional juz starts were split too. Table I gives the balance.

Quarters. With the hizb fixed and phrase starts only, a hizb holds about 18 phrases, 4.6 per quarter, and one long phrase can be a fifth of a quarter: the quarter CV is 9.6%, the worst quarter ±27.3%. Fig. 1 shows what splits could buy if every one proposed by the penalized solver were accepted: 105 bring the CV to 4.4%, near the 3.9% reached when every aya end is allowed (going lower would require cutting inside an aya). The review applies the meaning test to each: of 137 boundaries more than 3% off, with 465 candidates read, 60 were split at complete stops (e.g. 4:12, the share of spouses, which turns a −23% boundary into −3.6%) and 77 kept, because every closer point is a turn of a dialogue or story: Mūsā at the fire (28:29–35), Hūd (7:65–72), the messengers and their peoples (14:9–12). The result, CV 6.8% and worst ±20.8%, lies above the curve by the stops that meaning forbids, by the pause term of (2) (6.5% without it), and by one more rule set in review: no quarter starts at aya 2–7 of a surah; it opens the surah or starts at aya 8 or later (the traditional quarters break this twice). Juz 28 is exempt: its surahs of 11–24 ayat leave too few admissible starts. With the accepted splits, the rule costs the quarters 6.5% → 6.8% CV.

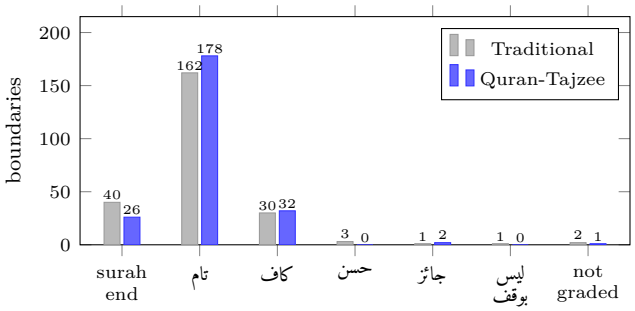


Figure 2: The pause before each of the 239 inner boundaries (hizb and quarter starts): the grade of the aya end before it in *al-Muktafā* and *Manār al-Hudā* (the stronger of the two).

VI. Discussion and limitations

The phrase test is ours. That no new boundary falls inside a phrase holds by construction; the phrases are one reading of the text, reviewed by one reader. A wording check independent of it, a first word that leans on the previous aya (a reply such as "قال" or a linking "ف", "ثم"), flags 21 traditional starts and a similar number of new ones, most of them a resumptive "ف" at a new subject ("فاصدع بما تؤمر"), but a few real: the 10-aya phrase limit splits long stories at turns of dialogue.

The classical grades. An independent judge is the pause grades of the *waqf* works. We extracted, from the OpenITI texts [9], the grade al-Dānī's *al-Muktafā* gives 1,844 aya ends and al-Ushmūnī's *Manār al-Hudā* 5,265 [6], [7]; where both grade an end they agree within one grade 96% of the time. Fig. 2 grades the aya end before every boundary. 99% of the new boundaries follow a surah end, a complete (تام) or a sufficient (كاف) stop, against 97% of the traditional ones. Since (2) uses these grades, the independent test is the mix-only objective (1) for the quarters, which knows nothing of them: it already reaches 96%, so the division drawn for balance at phrase starts is placed as well as the one drawn by stops. The pause term of (2) does this at a small cost: against the mix-only objective it moves 16 quarters, lowers the pause points from 520 to 493 and the starts graded below كاف from 9 to 3 (23:37, 26:116 (a hizb), 26:176), for a quarter CV of 6.5% → 6.8%. The juz openings inside a dialogue (18:75, 51:31) follow a sufficient and a complete stop: the classical grade asks for a new sentence, the phrase test for a new subject. Al-Ṣafāqusī's own note on the last word of each quarter was located for 233 quarters: 194 are the Madina ends. Of the other 39, he reports the Madina end as an alternative (of the Mashāriqa, the Maghāriba or "some") in 28; in 11 the Madina Mushaf departs from him.

Counting schools. The text is in the Kufan count, but al-Ṣafāqusī also asks whether a quarter's last word ends an aya in every count (فاصلة بلا خلاف). We check each boundary against the six systems of al-Dānī's *al-Bayān* [5],

[8]: 134 Kufan aya ends are not counted by some system. None of the 239 traditional boundaries follows one, which confirms the rule in the transmitted divisions. Disputed ends are therefore removed from $\mathcal{S}$: without that, the solver would start a quarter after one (20:42→20:49), at 6.8% → 6.8% CV; now 0 boundaries follow a disputed end.

Weights. Recitation and print agree so closely that their weights (50/50 for the hizb, $\frac{1}{3}$ each with the pause for the quarters) matter little; header lines of surahs, if counted in print, lengthen juz 30.

VII. Conclusion

Al-Ṣafāqusī's rule, that a quarter ends where a thought ends, is sound, but the transmitted quarters keep it only partly and vary more than twofold in length. Applying the rule to every boundary and choosing among the admissible stops by measured length gives hizb within ±3.6% of half their juz, and quarters with a CV of 6.8% against 11.7%, with no boundary inside a phrase, after a disputed aya end or, outside juz 28, at ayat 2–7 of a surah, and 99% of them at a stop the classical works grade complete or sufficient. The remaining unevenness is the price of not cutting a story or a dialogue, a choice the review tables make explicit for a human reviewer to keep or overturn.

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